

Modeling and Control Strategy for the Transition of a Convertible Tail-sitter UAV

J. Escareño, R.H. Stone, A. Sanchez and R. Lozano

Abstract—This paper addresses the problem of the transition between rotary-wing and fixed-wing flight of a tail-sitter unmanned aerial vehicle (UAV). A nonlinear control design is presented to regulate the vertical-flight dynamics of the vehicle. We present the dynamic and aerodynamic equations that model the behavior of the vehicle before (vertical flight), during and after (forward flight) the transition. A low-cost embedded system, including an homemade inertial measurement unit (IMU), is used to perform autonomous attitude-stabilized flight in vertical mode.

Index Terms—Tail-sitter, Backstepping, Embedded architecture, Inertial Measurement Unit.

I. INTRODUCTION

Tail-sitter UAVs have a number of advantages compared to other configurations. In comparison to conventional designs they pose much greater operational flexibility because they don't require a runway for launch and recovery but instead can operate from any small clear space. While other conventional designs partially overcome this limitation via the use of takeoff and landing aids such as catapults and parachutes, these all entail extra system complexity and logistic support. Although helicopter UAVs share the same operational flexibility as the tail-sitter, they suffer from well known deficiencies in terms of range, endurance and forward speed limitations due to the lower efficiency of rotor-born, rather than wing-born flight. Lastly, the other configurations that have been developed to archive the same goals as the tail-sitter, such as the tilt-wing, tilt-rotor and tilt-body, do so at the expense of significantly increased mechanical complexity compared to a tail-sitter that uses propeller wash over normal aircraft control surfaces to effect vertical flight control. By marrying the takeoff and landing capabilities of the helicopter with the forward flight efficiencies of fixed-wing aircraft in such a simple way, the tail-sitter promises a unique blend of capabilities at lower cost than other UAV configurations.

While the tail-sitter concept has great promise, it also comes with significant challenges. Foremost amongst these is designing controllers that will work over the complete flight envelope of the vehicle: from low-speed vertical flight through to high-speed forward flight. The main change in this respect (besides understanding the detailed aerodynamics) is the large variation in the vehicle dynamics between these

two different flight regimes. In vertical flight thrust is the dominant force and horizontal control is achieved via tilting the thrust vector, while maintaining attitude control using the propeller wash over the vehicle surfaces. (It should also be noted that as an extra complicating factor, the amount of wash and hence the control effectiveness of the control surfaces depends on the thrust generated). In horizontal flight the dominant force is the lift-force provided by the wings and horizontal control is achieved via tilting this lift vector (in banked turns), while again maintaining attitude control with the control surfaces in the free-stream flow. The design of a robust control system that can cover these two flight regimes and handle the non-linearities present in both is non-trivial task.

In [1] the author utilizes a LQR algorithm to control, in hover mode, the longitudinal-flight dynamics (attitude and position). In [2] the authors apply a PD control to regulate the pitch and yaw attitude, with a classical fixed-wing configuration, during vertical flight. In [3] we employ a saturation-based controller to stabilize the aircraft's position and attitude at hover flight, assuming that it is close enough to the origin so that we could ignore the nonlinearity present in the underactuated dynamics and also ignore the aerodynamics effects. This paper, however, considers the transition between rotary-wing and fixed-wing flight, and thus the algorithm (backstepping-based controller) is required to be robust enough to handle larger pitch angles away from the vertical. To do this the nonlinearity is explicitly accounted for, so that the UAV can achieve a vertical attitude from a considerable pitch angle, (for instance during the transition from horizontal back to vertical flight).

The paper is organized as follows: in Section 2, the mathematical model of the tail-sitter aircraft is presented. In Section 3, we develop a stabilizing control law for the vehicle in hover and forward flight mode. Simulations results are presented in Section 4. The experimental results are provided in Section 5. Conclusions and perspective are finally given in Section 6.

II. DYNAMICAL MODEL

The longitudinal model of vehicle that will be used in this paper is more complete than that used in previous papers about the tail-sitter [3]. This time simplified (though reasonable) treatment of the aerodynamic lift, drag and pitching moments will be given. It is important to consider these forces properly because they are fundamentally affected by the vehicle's motion and thus alter the basic dynamics involved. The analysis used will be based on combination

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of a low-order panel method aerodynamic model coupled with a simple actuator disc model of the flow induced by the propellers. In considering the aerodynamic forces on the vehicle it is essential to note that these are all associated with the flow induced by the propellers over the vehicle's aerodynamic surfaces as well as by the perturbation of this flow due to the motion of vehicle. Thus the starting point for any analysis is determination of the propeller slipstream velocities. In the following analysis the following assumptions will be made:

- A1.** Propeller Normal Forces are negligible;
- A2.** The slipstream velocity $V_{slip} \gg W$, the body normal velocity. This implies that the slipstream angle of attack is small;
- A3.** The vehicle aerodynamic surface are fully submerged in the propeller slipstream;
- A4.** Body aerodynamic forces are negligible;
- A5.** Drag forces on the vehicle are small compared to thrust and lift forces in hover flight, and small compared to lift during horizontal flight.

Assuming purely axial flow into the propellers, (a close enough assumption for the present case) simple actuator disc theory [see [1]] gives the induced far-slipstream axial velocity as:

$$V_{slip} = \sqrt{U^2 + \frac{2T}{A\rho}} \quad (1)$$

where U is the axial velocity of the vehicle, T , the thrust, A the total disc-area of the propellers and ρ the air-density. This velocity must now be combined with the relative normal velocity of the vehicle with respect to the air to obtain the aerodynamic forces. The relative normal velocity may be due to vehicle motion (with respect to still air) or due to wind impinging on a stationary vehicle or a combination of these. In the following derivation only the vehicle motion will be considered with the addition of wind an obvious extension. Fig. 1 shows the aerodynamic forces on a small UAV with its lifting surfaces fully submerged in the propeller slipstream. The forces consist of a lift force, L , perpendicular to the total flow vector, V_{tot} , a drag force parallel to V_{tot} , and a pitching moment, M , about the positive cartesian y axis, a drag force, D , parallel. For small angles of attack, the aerodynamic lift and moment forces will be proportional to the slipstream angle of attack, α , and to the deflection of the control surface, δ_e . The drag force variation is more complex, however, it will be neglected for the rest of this treatment as it will usually be significantly less than the lift and thrust terms. The above discussion can be summarized by:

$$C_l = C_{l_\alpha} \alpha_{slip} + C_{l_\delta} \delta \quad (2)$$

$$C_m = C_{m_\alpha} \alpha_{slip} + C_{m_\delta} \delta \quad (3)$$

where (2) and (3) are standard aerodynamic non-dimensional lift and moment coefficients defined as:

$$C_l = L / (\frac{1}{2} \rho V_{slip}^2 S) \quad (4)$$

$$C_m = M / (\frac{1}{2} \rho V_{slip}^2 S c) \quad (5)$$

In these equations S and c are a reference area and length respectively. Thus to obtain the lift and moment forces on the aircraft it is only necessary to obtain the slipstream axial velocity, which has already been done, the angle of attack and the aerodynamic parameters C_{l_α} , C_{l_δ} , C_{m_δ} , C_{m_α} which are a function of the geometry of the vehicle. To determine the angle of attack, it is necessary to obtain the W -component of relative velocity (the velocity component normal to the vehicle). This is given by:

$$W = \dot{X} \cos(\theta) - \dot{Z} \sin(\theta) - Q l_w \quad (6)$$

In this equation the first two terms come from the vehicle linear motion, while the third is due to rotary motion about the vehicle CG. This rotation at angular speed Q causes an effective linear motion at the lifting surface, which is displaced a distance l_w from the CG. This treatment is not entirely accurate in the case where the aerodynamic surface is large in comparison to l_w however, this small error will be ignored for simplicity. If the value of W is small in comparison to the slipstream velocity, (true under the small a-assumption), then $V_{tot} \approx V_{slip}$ and the angle of attack can be written as:

$$\alpha_{slip} = \arctan \left(\frac{\dot{X} \cos(\theta) - \dot{Z} \sin(\theta) - Q l_w}{V_{slip}} \right) \quad (7)$$

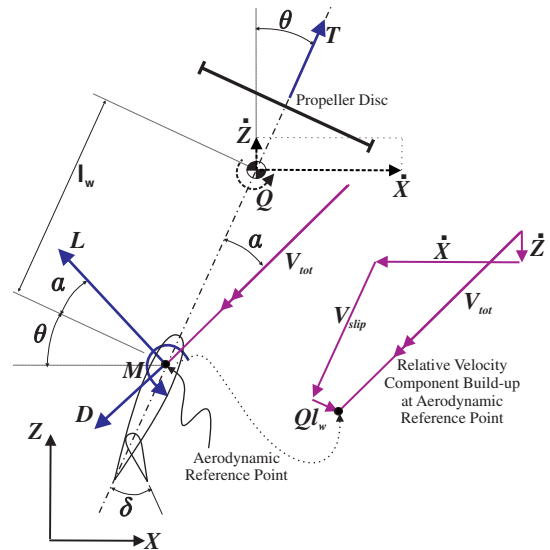


Fig. 1. Aerodynamic forces: a UAV with lifting surfaces fully submerged in the propeller slipstream

A. Equations of Motion (Cartesian Form)

The dynamical behavior of the tail-sitter vehicle [see figure 1] is described by the motion equations (8)¹

$$\begin{cases} m\ddot{X} = T \sin(\theta) - L \cos(\theta + \alpha) - D \sin(\theta + \alpha) \\ m\ddot{Z} = T \cos(\theta) + L \sin(\theta + \alpha) - D \cos(\theta + \alpha) - mg \\ I_{yy}\ddot{\theta} = -L \cos(\alpha)l_w + M \end{cases} \quad (8)$$

In these equations the lift, drag and moment terms can be represented as:

$$\begin{aligned} L &= \frac{1}{2}\rho S(C_{l_\alpha}\alpha + C_{l_{\delta_e}}\delta_e) \\ D &= \frac{1}{2}\rho S(C_{d_0} + KC_L^2)V_R^2 \\ M &= \frac{1}{2}\rho S c(C_{M_{\delta_e}}\delta_e)V_R^2 \end{aligned}$$

where ρ is the density; S is the wing area submerged in the flow; and c is the wing-chord.

B. Simplified model

For further control analysis let us consider a simplified model of (8). In this approximation we recall **A1**, **A4** and **A5**, and we consider the normalized values of the mass, gravity, the inertial mass moment (i.e. $g = 1, m = 1, I_{yy} = 1$). Therefore, (8) becomes

$$\begin{cases} \ddot{X} = T \sin(\theta) - L \cos(\theta + \alpha) \\ \ddot{Z} = T \cos(\theta) + L \sin(\theta + \alpha) - 1 \\ \ddot{\theta} = \frac{1}{I_{yy}}(-L_e \cos(\alpha)l_w) \end{cases} \quad (9)$$

Since the lift provided by elevator deflection δ_e mainly acts on the aircraft through the pitching moment, we may rewrite $\ddot{\theta}$ as:

$$\ddot{\theta} = u_\theta(\delta_e) \quad (10)$$

III. CONTROL STRATEGY

In order to observe the aerodynamical behavior during the vehicle's transition we have defined the following task: first, the UAV takes off vertically to a desired altitude z , meanwhile it regulates the attitude θ to reach a desired position x , afterwards, it switches to forward-flight modality controlling only the aircraft's attitude.

A. Vertical Flight Control

We consider the vertical flight as a critical stage for this vehicle, since our prototype's structure is oriented more towards a classical fixed-wing aircraft. In this flight regime exists two dynamic subsystems: the altitude dynamics, which is fully-actuated by the thrust T , whereas the $x - \theta$ dynamics is an underactuated subsystem. Recalling **A2**, we neglect the aerodynamical terms in the control design. However, we will show that our control law performs satisfactorily even if we include the aerodynamic term (perturbation) in the simulation study. The latter, allow us to rewrite (9) as:

$$\begin{cases} \ddot{X} = T \sin(\theta) \\ \ddot{Z} = T \cos(\theta) - 1 \\ \ddot{\theta} = u_\theta(\delta_e) \end{cases} \quad (11)$$

¹In these equations we have dropped the subscript "slip" from the angle of attack (α)

It is clear that (11a) can be stabilized with an feedback-linearizable input via the thrust T . An appropriate choice is

$$T = \frac{r_z + 1}{\cos(\theta)} \quad (12)$$

where, $r_z = -a_{z1}\dot{Z} - a_{z2}(Z - Z^d)$ with $a_{z1}, a_{z2} > 0$ and z_d is the desired altitude. From (12) we notice that the controller scope is restricted to

$$-\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

This interval is appropriate for vertical hover-mode, since the vehicle's operating point usually is more or less the horizontal orientation. However, it excludes the possibility for horizontal flight ($\theta = \frac{\pi}{2}$). This is entirely satisfactory as during horizontal flight the altitude dynamics are dominated by the wing lift force not the thrust force, which means that an alternative control strategy is required for that region of the flight envelope.

Substituting (12) in (11) we obtain

$$\begin{cases} \ddot{X} = (r_z + 1) \tan(\theta) \\ \ddot{Z} = r_z \\ \ddot{\theta} = u_\theta(\delta_e) \end{cases} \quad (13)$$

Note from (13b) that it exists a time large enough such that Z and $\dot{Z}$ are arbitrarily small, hence $x - \theta$ dynamics becomes:

$$\begin{cases} \ddot{X} = \tan(\theta) \\ \ddot{\theta} = u_\theta(\delta_e) \end{cases} \quad (14)$$

The state space representation of the previous equation is written

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = \tan(x_3) \\ \dot{x}_3 = x_4 \\ \dot{x}_4 = u_\theta \end{cases} \quad (15)$$

is easy to observe that (15) has a pure-feedback system. Therefore, we will employ the Backstepping technique to design a control law that stabilize the underactuated subsystem.

Firs step: Let us introduce a virtual state given by

$$z_2 = x_2 - x_2^v \quad (16)$$

at this step, we consider $z_2 = 0$ to isolate (15a) and it also implies that $x_2 = x_2^v$. This shows that the first order system is stabilisable through x_2 . To do so, we propose the Lyapunov function $V_1 = \frac{1}{2}x_1^2$. Thus, to render $\dot{V}_1$ negative definite we choose the following input

$$x_2 = x_2^v = -x_1 \quad (17)$$

which lead us to

$$\dot{V}_1 = -x_1^2 \quad (18)$$

Second step: Now, considering the case when $z_2 \neq 0$, the second order system (15a-15b) becomes²

$$\begin{cases} \dot{x}_{1a} = z_2 - x_1 \\ \dot{z}_2 = \tan(x_3) + x_2 \end{cases} \quad (19)$$

²We use the subindex a to denote the augmented state variable which contains the virtual state

At this step is convenient to introduce a second virtual state given by

$$z_3 = \tan(x_3) - x_3^v \quad (20)$$

considering $z_3 = 0$ implies that $\tan(x_3) = x_3^v$. Then, we use $\tan(x_3)$ to stabilize the augmented second order system (19). Note that, due to the $\tan(\cdot)$ nature (several equilibrium points), the controller acts within $-\frac{\pi}{2} < x_3 < \frac{\pi}{2}$, which is appropriate to perform vertical flight. We propose the Lyapunov function $V_2 = V_1 + \frac{1}{2}z_2^2$. Whose derivative $\dot{V}_2$ is rendered negative definite through the following control

$$\tan(x_3) = x_3^v = -z_2 - (x_1 + x_2) \quad (21)$$

leading $\dot{V}_2$ to

$$\dot{V}_2 = -x_1^2 - z_2^2 \quad (22)$$

Third step: We consider the case when $z_3 \neq 0$, turning out the following third order system

$$\begin{cases} \dot{x}_{1a} = z_2 - x_1 \\ \dot{z}_{2a} = z_3 - z_2 - x_1 \\ \dot{z}_3 = x_4 \tan^2(x_3) + 2 \tan(x_3) + 2x_2 \end{cases} \quad (23)$$

At this step is useful to introduce the third virtual state

$$z_4 = x_4 - x_4^v \quad (24)$$

with $z_4 = 0$, we obtain $x_4 = x_4^v$. To derive the controller for (23) we propose the Lyapunov function $V_3 = V_2 + \frac{1}{2}z_3^2$ whose derivative is rendered negative definite by

$$x_4 = x_4^v = -\left(\frac{z_3 + z_2 + 2 \tan(x_3) + 2x_2}{\tan^2(x_3) + 1}\right) \quad (25)$$

Fourth step: Finally, we consider $z_4 \neq 0$, turning out the following system

$$\begin{cases} \dot{x}_{1a} = z_2 - x_1 \\ \dot{z}_{2a} = z_3 - z_2 - x_1 \\ \dot{z}_{3a} = z_4(\tan^2(x_3) + 1) - z_3 - z_2 \\ \dot{z}_4 = u_\theta + \frac{\epsilon_1}{\tan^2 x_3 + 1} \end{cases} \quad (26)$$

with

$$\epsilon_1 = 3x_2 + 5 \tan x_3 + 3x_4(1 - \tan^2 x_3) - 2x_4 \tan x_3(3x_1 + 5x_2)$$

In order to deduce the controller for (26) we propose the Lyapunov function $V_4 = V_3 + \frac{1}{2}z_4^2$, whose derivative is definite negative as long as we apply the following

$$u_\theta = -z_4 - \frac{\epsilon_2}{\tan^2 x_3 + 1} \quad (27)$$

with

$$\epsilon_2 = 2x_1 + 5x_2 + 6 \tan x_3 + 3x_4(1 - \tan^2 x_3) - 2x_4 \tan x_3(3x_1 + 5x_2)$$

The virtual states obtained along the controller design are given by:

$$\begin{aligned} z_2 &= x_2 + x_1 \\ z_3 &= \tan(x_3) + 2x_1 + 2x_2 \\ z_4 &= x_4 + \frac{3x_1 + 5x_2 + 3 \tan(x_3)}{\tan^2(x_3) + 1} \end{aligned}$$

The final control law (27) stabilizes the underactuated subsystem (15), at hover mode, in attitude and position. It is worth to remark that we have used unitary gains during the backstepping design to avoid the abuse in the notation.

TABLE I
SIMULATION'S PARAMETERS

PARAMETERS	VALUE
m	1kg
g	1m/s ²
I _{yy}	0.01889kg · m ²
l _w	0.3m
ρ	1.225kg/m ³
C _{lα}	1rad ⁻¹
S	0.1653m ²
S _e	0.01m ²
a _{θ1}	10
a _{θ2}	10
a _{z1}	1.5
a _{z2}	1

B. Forward flight control

The remaining stage in the assignment previously outlined, is the control of the pitch attitude during forward flight. Then, to achieve this goal we apply the following control input:

$$u_\theta = -a_{\theta_1}\dot{\theta} - a_{\theta_2}(\theta - \theta^d) \quad (28)$$

with $a_{\theta_1}, a_{\theta_2} > 0$. At this stage, the convertible UAV is considered as a classical fixed-wing vehicle. The translational behavior of the vehicle includes the altitude and the horizontal motion, whose control inputs are, respectively, the elevator deflection and the thrust.

IV. SIMULATION STUDY

In order to validate the control strategy described in Section 3, we have run simulations (see table I) to observe the performance of the aircraft in both flight regimes. The UAV starts performing hover flight from the initial position $(x, z, \theta) = (0, 0, \frac{\pi}{8})$, then the vehicle reaches a desired altitude ($z = 2$) with desired position ($x = 0.6$) and pitch attitude ($\theta_d = 0$), then, at the time $t = 25$ a perturbation ($\theta_p = \frac{\pi}{8}$) in the attitude is applied. At the time $t = 50$ the aircraft switches to forward flight, considering $\theta_d = \frac{\pi}{4}$. Finally, at the time $t = 60$ a perturbation $\frac{\pi}{8}$ is applied to the pitch attitude.

The simulation corresponds to the system described by the equations (9). Therefore, we notice in the figure (2) that the hover-mode controllers (12) and (27) are robust enough to cope with the aerodynamical perturbation (adverse lift) created by a considerable vertical position deviation.

In figure (3) we observe that the elevator deflection presents an overdamped behavior, mainly due to the elevator's operation interval and the unitary gains used in the backstepping algorithm. Moreover, we can observe the thrust performance in both modalities. In figure (4) we observe the variation of the lift during the presence of a perturbation in the pitch attitude. After the transition we observe the relation between the lift-force the attack angle.

V. EXPERIMENTAL RESULTS

The configuration of the tail-sitter UAV is a compromise between rotary and fixed wing aircraft. Hence, the flight

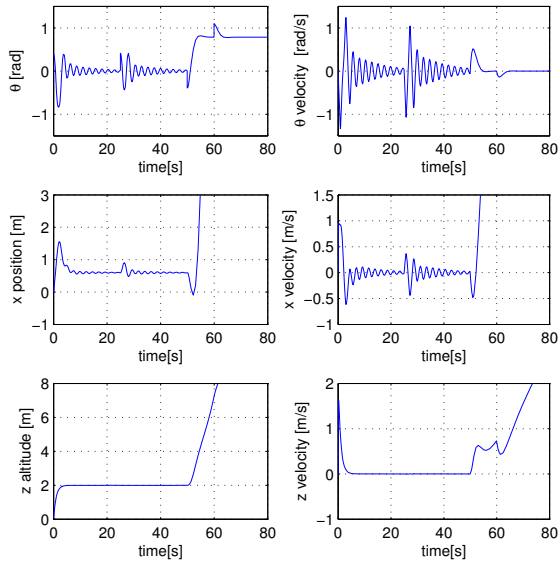


Fig. 2. UAV's states

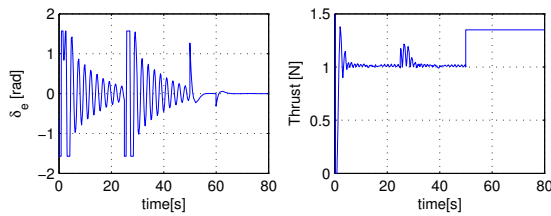


Fig. 3. UAV's control inputs

control of the UAV, in both operations modes, depends on dynamic and aerodynamic terms. The altitude of vehicle is regulated by increasing or decreasing the propeller thrust. The roll torque is obtained from the difference of the rotors angular velocities. Since the control surfaces are submerged in the propeller slipstream (prop-wash), the aerodynamic forces are generated by the elevator and ailerons deflections to provide the pitch and yaw motion, respectively.

A. Embedded system

In this section we described the embedded system to perform an autonomous attitude-stabilized flight. The system is composed by two main modules: the inertial measurement unit (IMU), the embedded control module.

- 1) *Inertial Measurement Unit:* We have built an inertial measurement unit (IMU) that includes a dual-axis accelerometer, which senses the angular position of the vehicle (ϕ , θ), and three gyros, which sense the angular rate of the vehicle ($\dot{\phi}$, $\dot{\theta}$, $\dot{\psi}$). The yaw³ angle is obtained by the integration of yaw angular rate.
- 2) *Embedded Control:* The IMU (analog signals) feeds the PIC microcontroller which sends this information

³The associated drift is tackled by resetting the yaw angle at a certain drift value

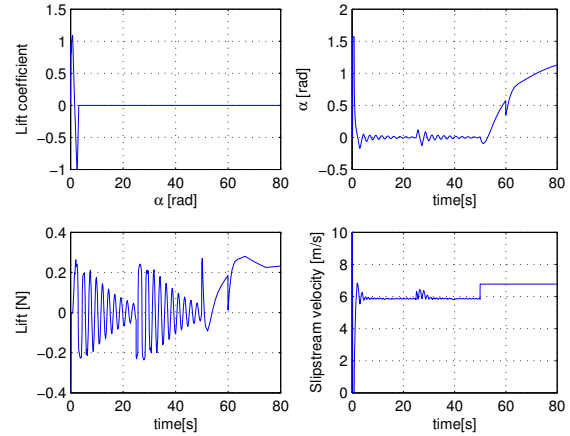


Fig. 4. UAV's aerodynamics

to the Rabbit microcontroller through the serial port. The inertial information is filtered to get rid of the electrical and mechanical noise (mainly due to rotor's gearbox). Finally, the control signal is sent to the motors (propellers) and servomotors (ailerons-elevator) via the PWM ports (see fig. 7).

B. Attitude performance

So far, we have performed experimental test, on the prototype previously described, to stabilize the aircraft's attitude and in vertical flight mode. The performance of this experiment is shown in figures 8, 9 and 10. y

VI. CONCLUDING REMARKS

In this paper we have presented a control strategy to perform the transition of a tail-sitter UAV from vertical to forward flight mode. We have presented in detail the longitudinal model of the aircraft as well its simulation in order to observe the dynamic and aerodynamic performance during the transition. For vertical flight mode, we have derived control algorithm, based on backstepping, robust enough to cope with considerable aerodynamical perturbations. An embedded autopilot was successfully tested for the attitude stabilization in vertical flight.



Fig. 5. VTOL prototype

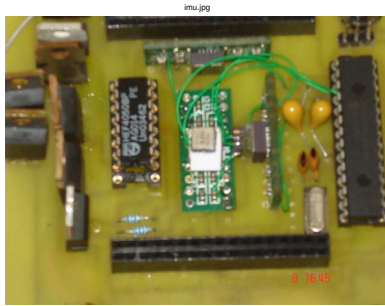


Fig. 6. Homemade IMU

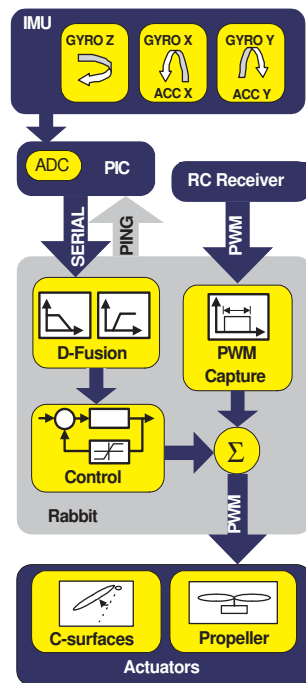


Fig. 7. Homemade IMU

The challenge is now to expand the control strategy for the 6-DOF nonlinear model of the tailsitter UAV for both operation regimes. Moreover, the improvement of the homemade IMU in such a way that it can be capable to measure the inertial attitude in the whole vehicle's operation range, as well the incorporation of sensors such as GPS or vision-based sensors.

REFERENCES

- [1] R. Hugh Stone, "Control Architecture for a Tail-sitter Unmanned Air Vehicle", *5th Asian Control Conference*, Melbourne, Australia, July 23-25, 2004.
- [2] William E. Green and Paul Y. Oh, "Autonomous Hovering of a Fixed-Wing Micro Air Vehicle", *International Conference on Robotics and Automation*, Orlando, Florida, USA, May, 2006.
- [3] J. Escareno, S. Salazar and R. Lozano, "Modeling and Control of a Convertible VTOL Aircraft", *45th IEEE Conference on Decision and Control*, San Diego, California, December 13-15, 2006.
- [4] H. Stone, "Aerodynamic Modeling of a Wing-in-Slipstream Tail-Sitter UAV", *Biennial International Powered Lift Conference and Exhibit*, Williamsburg, Virginia, Nov. 5-7, 2002.
- [5] H. Goldstein, C.P. Poole and J.L. Safko, *Classical Mechanics*, Addison-Wesley Publishing Company, Inc., Massachusetts, 1983.
- [6] A. Bedford, and W. Fowler, *Dynamics*, Addison-Wesley Publishing Company, 1989.
- [7] P. Castillo, R. Lozano A. Dzul, "Modeling and control of mini flying machines", Springer-Verlag, July 2005.
- [8] R. Lozano, et. al, *Dissipative Systems Analysis and Control*, Springer-Verlag, 2001
- [9] I. Fantoni and R. Lozano, *Nonlinear Control for Underactuated Mechanical Systems*, Springer-Verlag, 2002.
- [10] B. Etkin and L. Reid, *Dynamics of Flight*, J. Wiley & Sons, Inc., 1991.
- [11] B. L. Stevens and F.L. Lewis, *Aircraft Control and Simulation 2ed.*, J. Wiley & Sons, Inc., 2003.

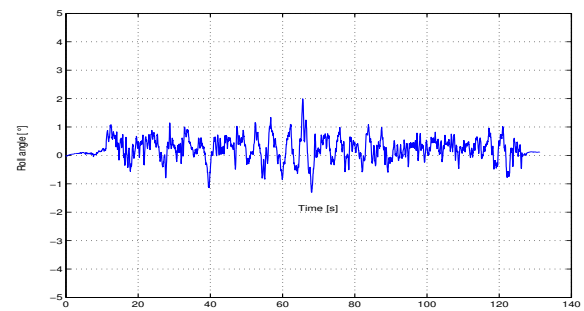


Fig. 8. Roll Angle performance

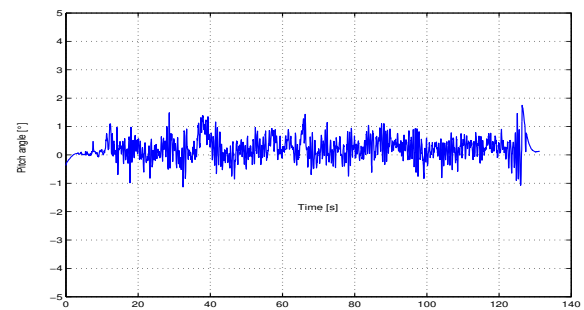


Fig. 9. Pitch Angle performance

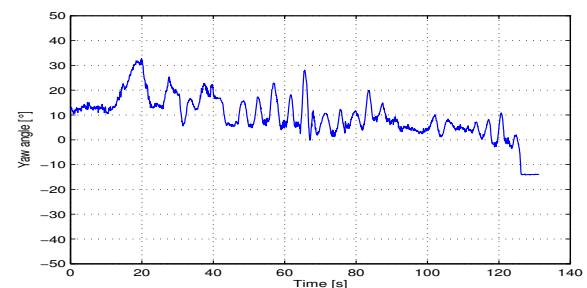


Fig. 10. Yaw Angle performance